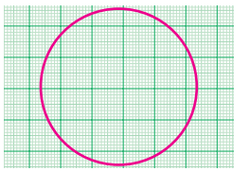
Activity

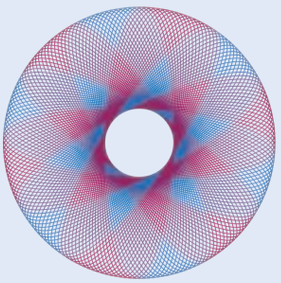
Draw circles of different measurements on a graph sheet. Find the area by counting the number of squares enclosed by the circle. So we get a rough estimate of the area of the circle.



Designs obtained by spirograph.

Shown below are some shapes using a spirograph. We can observe that each of the different designs are circular.





Think

If the area and circumference of the circle are equal then what will be the value of the radius?

Example 2.9 Find the area of the circle of radius 21 *cm* (Use $\pi = 3.14$).

Solution

$$\text{Radius } (r) = 21 \text{ cm}$$

$$\text{Area of a circle} = \pi r^2 \text{ sq. units}$$

$$= 3.14 \times 21 \times 21$$

$$= 1384.74$$

$$\text{Area of the circle} = 1384.74 \text{ cm}^2$$

Example 2.10 Find the area of a hula loop whose diameter is 28 *cm* (use $\pi = \frac{22}{7}$).

Solution

$$\text{Given the diameter } (d) = 28 \text{ cm}$$

$$\text{Radius } (r) = \frac{28}{2} = 14 \text{ cm}$$

$$\text{Area of a circle} = \pi r^2 \text{ sq.units}$$

$$= \frac{22}{7} \times 14 \times 14$$

$$\text{So, the area of the circle} = 616 \text{ cm}^2.$$



Fig. 2.20

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